

Bachelor's Thesis

Predicting interactions between phages and bacteria through proteome representation

A latent space approach across CRC, IBD and GvHD cohorts

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Abstract

Bacteriophages shape the composition of the human gut microbiome, but which phage infects which bacterium is rarely observed directly. Two proxies can be derived from sequencing data: a CRISPR spacer matching a viral genome, which records a past infection of the bacterial lineage, and a co-abundance edge estimated by a graphical lasso. This thesis predicts both of the proxies from the same representation, in which bacterial genera and viral operational taxonomic units are described only by protein clusters formed *de novo* within each cohort. No functional annotation is used. Additionally, we train and evaluate each model on the identical set of genus–vOTU pairs, on identical folds, in three independent cohorts, and ask whether one latent representation can serve both.

The two targets behave very differently. CRISPR linkage reaches 0.78–0.81 AUC across cohorts, while the co-abundance edge is predicted consistently above chance but far more weakly, at 0.55–0.61, with negative R^2 throughout for regression on the continuous edge probability. The weak signal does not improve with a change in the model, representation, binarisation threshold or taxonomic resolution. A joint model trained on both targets does not unify them. Even though it results in a small gain for the co-abundance task, it has no clear benefit for the CRISPR task. Additionally, the small gain survives even when the CRISPR labels are permuted, which implies that the CRISPR task acts as a regulariser for the co-abundance task rather than as a source of shared signal. The one component of this pipeline carrying explicit biological information is the CRISPR target; where it entered, the signal is strong, and where it did not, only a weak one remains.

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1 Introduction

1.1 Motivation and overview

The human gut holds a dense bacterial community alongside a large population of the viruses that infect it, most of them bacteriophages (Shkoporov and Hill, 2019). Infection requires a match between molecular structures on both partners, so a phage generally infects only a narrow range of hosts, and in removing some bacteria and not others it shapes the community. Shifts in that composition have been associated with colorectal cancer, inflammatory bowel disease and graft-versus-host disease (Wirbel et al., 2019, Lloyd-Price et al., 2019, Peled et al., 2020). Which phage infects which bacterium is therefore worth knowing, and is rarely observed directly.

Two proxies are available from sequencing data, and they are of different kinds. A CRISPR spacer is a fragment of viral DNA stored in the bacterial genome, so a spacer matching a viral genome shows that the bacterium’s lineage was infected by that virus at some point in the past; the match itself is found computationally (Zhang et al., 2021). A co-abundance network, on the other hand, is estimated statistically: a graphical lasso over abundances measured across samples marks pairs whose abundances covary once every other organism is held fixed (Friedman et al., 2008, Liu et al., 2010).

Computational host prediction is an established problem, but existing approaches build their features from curated references and assign a host taxon to a phage (Edwards et al., 2016, Coutinho et al., 2021, Roux et al., 2023, Nie et al., 2024). This thesis describes organisms only by protein clusters formed *de novo* within each cohort, without functional annotation, and predicts both proxies on the identical set of genus–vOTU pairs in three cohorts. Since the two targets are meant to describe one phenomenon, it also asks whether a single latent representation can serve both.

1.2 Research questions and outline

The thesis addresses three questions.

1. Can CRISPR linkage between a bacterial genus and a vOTU be predicted from an annotation-free protein-cluster representation of the two organisms?
2. Can a co-abundance edge between the same pair be predicted from the same representation?
3. Does a latent representation trained on both targets serve either better than one trained on a single target?

The contribution is a controlled comparison: both proxies are predicted from identical features, on identical folds, over the identical set of pairs, in three independent cohorts, so any difference between them is a difference between the targets and not between the data or the models.

Figure 1 summarises that setup. Each bacterial genus and each vOTU is described by its own annotation-free protein-cluster profile; the two profiles are combined into a single pair representation, and one model maps that representation to both targets.

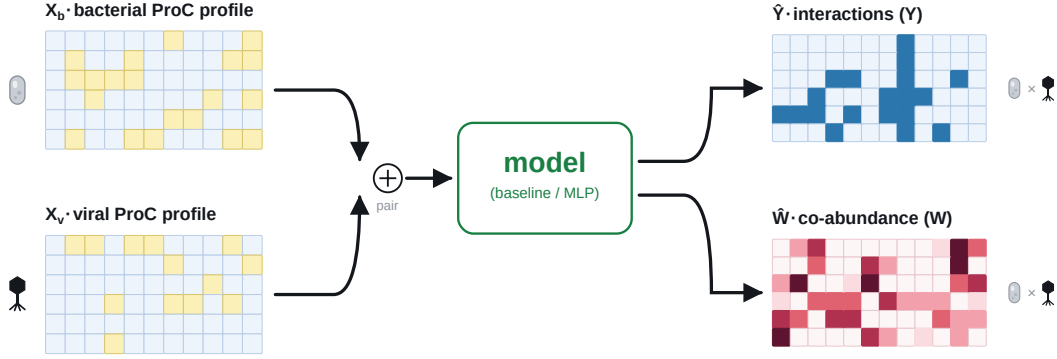


Figure 1: Overview of the prediction setup. A bacterial genus and a vOTU are each described by an annotation-free protein-cluster (ProC) profile, X_b and X_v . The two profiles are combined into one pair representation, which a single model maps to both targets: CRISPR linkage $\hat{Y}$ and the glasso co-abundance edge $\hat{W}$. Both targets are predicted from the identical representation, on identical folds, over the identical set of genus–vOTU pairs. This figure was created by Claude Opus 5 (Anthropic).

2 Background

2.1 The phage–bacteria interaction

A microbiome is the community of microorganisms living in a particular environment, together with their genetic material, interactions, and often the surrounding ecological context. They can be found in soil, in water, and on and inside multicellular organisms such as humans. One of the examples of a microbiome is the gut microbiome. The gut microbiome contains a dense bacterial community alongside a large population of viruses that infect some of the bacteria inside the microbiome (Shkoporov and Hill, 2019).

Some of those viruses are bacteriophages (also called phages), meaning viruses that are hosted by bacteria. They make up the majority of the gut virome (Shkoporov and Hill, 2019). A phage consists of little more than a genome enclosed in a protein capsid, in many cases attached to a tail. Infection proceeds in a fixed order. First, the proteins at the tip of the tail bind a specific molecular structure on the surface of the bacterium, anchoring the phage to the cell. Then, the genome of the phage passes through the tail into the bacterium while the capsid remains outside. Once inside, the genome is either replicated with the machinery of the host, which assembles new phages and destroys the cell as they are released, or it is inserted into the bacterial chromosome, where it is copied along with the host genome at every cell division until a later stimulus returns it to the first route. Because the initial binding requires a match between two particular molecular structures, a phage can generally infect only a small number of bacterial species, which makes the interaction host-specific.

Phages therefore act continuously on bacteria, removing some members and leaving the rest untouched. The function of the microbiome as a whole depends on its composition, and since the composition changes with bacteriophages, its function changes as well. These changes in the composition of the gut microbiome have been associated with a range of

conditions, including colorectal cancer (CRC), inflammatory bowel disease (IBD) and graft-versus-host disease (GvHD) (Wirbel et al., 2019, Lloyd-Price et al., 2019, Peled et al., 2020).

2.2 Protein clusters

Proteins are essential to the interaction between phages and bacteria, as everything from the tail of the phage that connects to the bacterium to the systems that defend the bacterium against the phage is made of proteins. The proteins of an organism therefore describe whether it is able to take part in an infection.

One of the major problems with using the proteins themselves is that even a single genome carries thousands of proteins, and that the same protein is rarely exactly the same sequence in two organisms. Two organisms that carry the same kind of tail protein would therefore be described by two different entries, and no vocabulary shared between organisms would exist. This is solved by grouping protein sequences that are similar enough to each other into protein clusters, where every cluster stands for a protein family rather than for one particular variant of it. MMseqs2 (Steinegger and Söding, 2017) performs this grouping systematically for large collections of sequences. At that size, it is not feasible to compare each sequence to all other sequences, so it first searches for short matching subsequences to find candidate pairs, scores those cheaply, and aligns only the ones that survive, before forming the clusters from the remaining similarities.

After each protein is assigned to a cluster, an organism can be written as a vector that counts how many of its proteins fall into each cluster. Collecting these vectors gives the feature matrices X_b for the bacteria and X_v for the viruses, with the organisms as rows and the clusters as columns. Since any single organism carries only a small share of all clusters that occur across the data, the features are high-dimensional and mostly zero. Their rows are also genera and vOTUs rather than single genomes, so a profile describes the repertoire of a whole taxonomic unit and not that of one organism.

2.3 CRISPR and SpacePHARER

CRISPR is an adaptive immune system found in many bacteria. A CRISPR array is a region of the bacterial genome built from short repeated sequences that are separated by unique sequences called spacers, together with a set of neighbouring *cas* genes (Hille et al., 2018). The system operates in three stages. When foreign DNA enters the cell, proteins encoded by the *cas* genes cut a short fragment out of it and insert that fragment into the array as a new spacer. Then, that array is transcribed and processed, so that every spacer ends up in a small RNA molecule serving as a guide. If DNA matching one of those guides enters the cell again, it is cut and the infection fails. Each spacer is therefore a record of an infection that the bacterial lineage encountered in the past, written into its own genome.

The CRISPR mechanism implies that if the spacers of bacteria are compared against the genomes of phages, one can determine that the bacterium, or an ancestor of it, was infected, while the absence of a match does not establish the opposite. One of the main issues is that spacers are only around 30 nucleotides long, meaning they are short enough

that a standard nucleotide search loses sensitivity, and a spacer that was acquired long ago will have diverged from the phage genome it was taken from. SpacePHARER (Zhang et al., 2021) addresses both. It translates the spacers and the phage genomes in all six reading frames and compares them at the protein level, where divergence is tolerated better, before realigning the resulting hits at the nucleotide level. It then combines the evidence from all spacers of one bacterium against one phage genome into a single score, so that several weak matches can be significant together, and reports only the pairs with a false discovery rate below 0.05.

There are further caveats to those pairs. Only bacteria that carry a CRISPR system can produce spacers at all, and the carriage differs between taxonomic groups. Spacers are also lost over time, so an interaction that left no surviving record cannot be distinguished from one that never happened. The record is historical rather than current, and spacers are acquired from mobile genetic elements in general and not from phages alone. The reported pairs are therefore best read as a lower bound of the interactions that happened.

2.4 Graphical lasso and StARS

Another signal for phage–bacteria interaction can be obtained from abundances rather than mechanistic evidence. If the abundances of a bacterium and a phage are measured across a set of samples, and the two organisms interact, then their abundances should vary together to some degree. Pairs whose abundances covary can therefore be treated as candidate edges of a co-abundance network. The problem with using the correlation between two organisms is that it also reacts to associations that pass through a third one. For example, if two bacteria both increase under the same condition, and a phage follows one of them, the phage will correlate with the other bacterium as well, without any relation between the two. Therefore, a correlation network is dense with indirect edges.

The density problem is solved by calculating the correlation between two organisms while holding all the other organisms constant. This is described by the inverse of the covariance matrix, which is called the precision matrix Θ . An entry of Θ that is zero means that the two organisms are conditionally independent given all others, so a nonzero entry is a direct association rather than one inherited through a third organism. However, estimating Θ by inverting the sample covariance is not possible when there are more organisms than samples. The graphical lasso (Friedman et al., 2008) solves this by adding an L_1 penalty to the Gaussian log-likelihood,

$$\hat{\Theta} = \arg \min_{\Theta \succ 0} \left\{ -\log \det \Theta + \text{tr}(\hat{\Sigma}\Theta) + \lambda \|\Theta\|_1 \right\}, \quad (1)$$

which forces most entries of $\hat{\Theta}$ to be exactly zero. The penalty weight λ controls how many edges remain.

This leaves λ to be chosen. Since the network is not predicting anything, there is no prediction error to cross-validate, and the usual likelihood-based criteria tend to select too many edges. Instead, StARS (Liu et al., 2010) chooses λ via stability. The samples are subsampled B times, the graphical lasso is fitted on each subsample, and λ is selected as the least sparse value at which the disagreement between the subsampled edge sets stays below a threshold β . The accepted instability is controlled by the new threshold,

meaning that a larger β returns a denser network and a smaller one a sparser network. As a result, it gives a score for each pair, namely the fraction of the subsamples in which its edge was selected,

$$W_{ij} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}[\hat{\Theta}_{ij}^{(b)} \neq 0], \quad (2)$$

which lies between zero and one. Taking only the rows belonging to bacteria and the columns belonging to viruses gives a bipartite matrix of the same shape as the CRISPR one.

W_{ij} is not an observation and not a biologically measured quantity, but the output of an estimator applied to abundance data and a selection frequency rather than a strength of association. Two organisms may covary for reasons that have nothing to do with infection, and an edge that is selected in most subsamples is evidence that the estimator is stable, not that the two organisms interact. This caveat is not specific to these data: correlation-based methods recover known infection networks at close to chance level in simulated virus–microbe communities (Coenen and Weitz, 2018), and comparable cautions have been raised for microbial association networks in general (Pinto et al., 2022) and for reading viral ecology off abundance correlations in marine metagenomes (Alrasheed et al., 2019).

2.5 Related work

Predicting which phage infects which bacterium is an established computational problem, and most existing approaches work from genome sequences: they match CRISPR spacers, search for phage sequences inside bacterial genomes, or compare the sequence composition of the two organisms, and they are usually evaluated against reference collections in which the host of a phage is already known (Edwards et al., 2016). A separate line of work infers association networks from abundance data, where an edge is a statistical association whose ecological meaning is not settled (Friedman et al., 2008, Liu et al., 2010). This thesis sits between the two. It uses neither a curated reference collection nor any functional annotation, but describes the organisms by protein clusters formed within each cohort and treats CRISPR linkage and co-abundance as two separate prediction targets on the same features, within three disease cohorts.

Two recent surveys place this more precisely. Existing methods have been sorted by the features they use, covering CRISPR spacers, prophage and homology evidence, k-mer composition, viral marker genes and receptor-binding proteins (Nie et al., 2024), and benchmarked against reference collections in which the host of each virus is already annotated (Shang et al., 2025). The closest prior work is RaFAH (Coutinho et al., 2021), which predicts a host from the protein content of a virus. A second close comparison is iPHoP (Roux et al., 2023), which integrates several such signals for metagenome-derived viruses. Both build their features from curated resources and both return a host taxon for a virus. The setting here differs in two respects: the protein clusters are formed *de novo* within each cohort and carry no functional annotation, and a prediction is made for every genus–vOTU pair on a full grid rather than as a taxon assignment. We are not aware of prior work that combines the two.

3 Data

3.1 Cohorts

The data come from three published studies of the human gut microbiome, covering colorectal cancer (CRC) (Wirbel et al., 2019), inflammatory bowel disease (IBD) (Lloyd-Price et al., 2019) and graft-versus-host disease (GvHD) (Peled et al., 2020). The clinical endpoints of the studies do not play a role in this thesis. The contributions of the studies are a set of bacterial genera, a set of viral operational taxonomic units (vOTUs), and the abundance and protein-cluster profiles of both, so that a cohort enters the analysis as the grid of candidate phage–bacteria pairs summarised in Table 1.

Table 1: Cohorts and units of analysis

	CRC	IBD	GvHD
Samples	84	148	90
Bacterial genera	75	94	95
vOTUs	186	293	379
Candidate pairs	13,950	27,542	36,005

We applied the same analysis to three different cohorts rather than one, because a result obtained in a single cohort cannot be separated from the properties of that cohort. Repeating the same analysis in three datasets collected by different groups and in different populations turns a finding that holds in all three into a statement about the signal rather than about one dataset.

The cohorts do not share the same feature vocabulary. The protein clusters are derived within each cohort separately, so a model fitted in one cohort reads columns that do not exist in another and cannot be transferred to it. Every model in this thesis is therefore trained and evaluated inside a single cohort, and the comparison between cohorts is a comparison of findings and not of transferred predictions.

3.2 Protein-cluster representation

Both bacteria and viruses are described in the same way without any annotation database. The proteins predicted from all genomes of a cohort are clustered *de novo* with MMseqs2 (Steinegger and Söding, 2017), and a genus or vOTU is then a profile over those clusters. Nothing about the function of a cluster enters the representation, and because clustering is repeated per cohort, the three cohorts share no feature vocabulary.

Bacteria and virus protein features exhibit different sizes. For example, in the CRC cohort, the bacterial genomes yield 70,625 clusters, a median of 3,765 per genus, whereas the viral genomes yield 2,756 clusters, a median of 30 per vOTU. Selection therefore keeps every viral cluster and cuts the bacterial matrix to the same width by variance, discarding 90 to 96 per cent of it (Table 2, Figures 18 and 16). We score variance on raw counts so that the selection is independent of the transform applied afterwards; which transform, and whether the halves are concatenated or summed, is taken up in Section 4.3.

Table 2: Protein clusters before and after the variance filter

	CRC	IBD	GvHD
Bacterial clusters	70,625	78,517	72,646
kept	2,756	4,226	7,168
Viral clusters	2,756	4,226	7,168
kept	all	all	all
Pair vector	2,756	4,226	7,168

3.3 Constructing the bipartite grid

The rows of the grid are bacterial genera and the columns are viral operational taxonomic units, indexed by i and j respectively. Because a genus or a vOTU without features cannot be passed to the model, we only take genera and vOTUs that appear in the CRISPR table and also carry a protein-cluster profile (Steinegger and Söding, 2017). The grid is then the full Cartesian product of the two surviving sets, so that every genus is paired with every vOTU and the unit of analysis is the pair rather than the organism. This is what produces the pair counts in Table 1. In IBD we dropped one genus since it had no protein-cluster profile, whereas all genera were kept in the other two cohorts.

We place the CRISPR target on the grid directly as it is indexed by the same genera and vOTUs. It is used in binary form, so that $Y_{ij} = 1$ if at least one spacer of genus i matches vOTU j (Zhang et al., 2021) and $Y_{ij} = 0$ otherwise; the number of matching spacers is discarded.

The glasso matrix is estimated separately from the abundance data (Friedman et al., 2008, Liu et al., 2010), from $B = 50$ subsamples at an instability threshold of $\beta = 0.2$; a sparser network at $\beta = 0.1$ was also available and is not used here. It is estimated on its own set of entities, with its own naming convention, meaning it does not necessarily align to the grid in its raw form. We normalise the names on both sides before any matching, by lowercasing, removing every character that is not a letter or a digit, and then reindex W onto the grid. We then fill the cells that have no genus or vOTU counterparts in the glasso matrix with zero, and record a second matrix of the same shape marking which cells were matched. That matrix is the observation mask.

The glasso matrix is continuous. $W_{ij} \in [0, 1]$ is the fraction of subsamples in which the edge between genus i and vOTU j was selected, so it is a selection frequency rather than a binary statement about the pair. During the pipeline, we use it in two forms. In the regression task we keep the value itself, and in the classification task we threshold it at zero, so that the label is 1 if $W_{ij} > 0$ and 0 otherwise. Together with the CRISPR target this gives the three prediction targets used throughout this thesis, referred to as y , w_{reg} and w_{class} .

3.4 The observation mask

The glasso matrix does not cover the whole grid. It is estimated on the entities that survive the prevalence filter of the abundance data, which means it is not the same set as the entities that carry both a CRISPR entry and a protein-cluster profile. Thus, only some of the genera and only some of the vOTUs of the grid have a counterpart in W . We

record this in a binary matrix M of the same shape as the grid, with $M_{ij} = 1$ if genus i and vOTU j were both matched and $M_{ij} = 0$ if not. A cell is matched exactly when its row and its column are both matched, so M is the outer product of the two indicator vectors and the observed cells form a block rather than a scattered pattern. In CRC, 62 genera and 93 vOTUs are matched, which gives $62 \times 93 = 5,766$ observed pairs, or 41% of the grid. The corresponding figures are $75 \times 70 = 5,250$ pairs (19%) in IBD and $56 \times 344 = 19,264$ pairs (54%) in GvHD.

Even though the unmatched cells are stored as zero on the grid, that zero is not a statement about the pair. Treating it as an observed absence would teach the model that unmatched W edges mean absence. In reality the cell is unobserved, so it is not known from the data if the interaction was absent or present. We therefore exclude those cells entirely from the pipeline.

Because Y is defined on the whole grid, the intersection between the Y -observed and the W -observed set is the W -observed set itself, and this single mask places all three tasks on the same pairs. The whole pipeline is applied to an identical set of pairs, meaning a difference between the Y task models, W task models and joint models is a difference between the targets and not between the data they were shown.

The cost is that Y is no longer evaluated on the full grid. In CRC, 5,766 of the 13,950 pairs remain and the CRISPR prevalence rises slightly from 3.1% to 3.4%, so the Y results describe the masked subset and not the complete genus $\times$ vOTU grid.

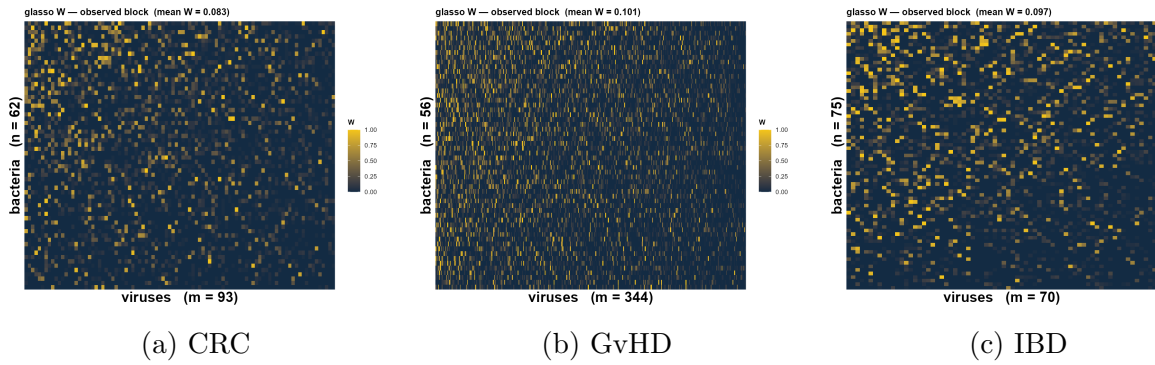


Figure 2: Glasso matrix on the observed block only, by cohort. Every cell carries a real W value, so no unobserved region remains; n is the number of bacterial genera and m the number of vOTUs.

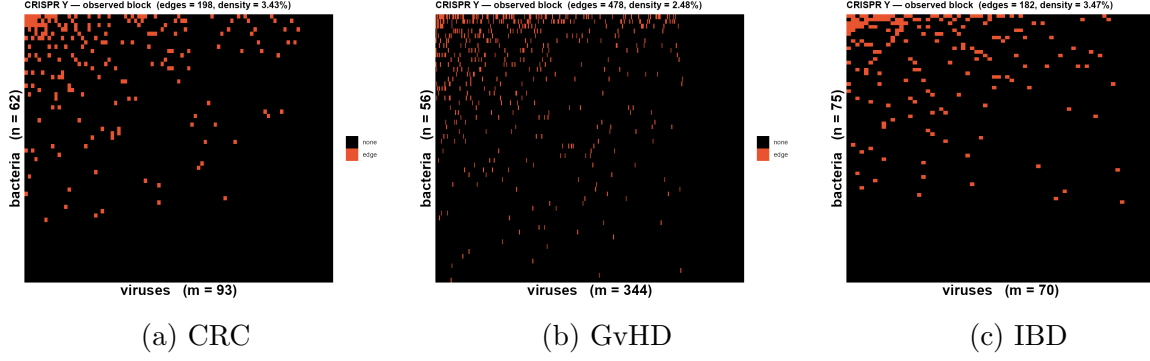


Figure 3: CRISPR matrix on the observed block only, by cohort, on the same pair set as Figure 2; n is the number of bacterial genera and m the number of vOTUs.

3.5 Descriptive statistics

Unless stated otherwise, the numbers in this subsection refer to the masked set, since that is the set every model is trained and evaluated on. Table 3 collects them. The two tasks differ greatly in terms of their positive classes. CRISPR linkage marks between 2% and 4% of the observed pairs, while a glasso edge is present in roughly two out of five, which means a model faces a strong imbalance on Y and a nearly balanced one on W . The class balance on the W task is a consequence of the thresholding rather than a property of the network. W is a selection frequency, so $W > 0$ admits every pair that was selected in at least one of the fifty subsamples, whereas the glasso network at the selected penalty has a bacteria–virus density of only 2% to 4% across the three cohorts.

Table 3: The masked set, by cohort

	CRC	IBD	GvHD
Observed pairs	5,766	5,250	19,264
CRISPR positive	198 (3.4%)	182 (3.5%)	478 (2.5%)
Glasso edge ($W > 0$)	2,473 (42.9%)	2,019 (38.5%)	9,451 (49.1%)
Mean W	0.083	0.097	0.101
Both	84	92	265

The two targets also overlap far less than their prevalences alone would suggest. If CRISPR linkage and a glasso edge were independent of each other, the expected number of pairs carrying both would be

$$\mathbb{E}[\text{both}] = \frac{n_Y n_W}{n_{\text{obs}}}, \quad (3)$$

with n_Y and n_W the number of positives of each target and n_{obs} the number of observed pairs. In CRC this gives $198 \times 2,473/5,766 = 85$, against the 84 pairs that actually carry both. The corresponding figures are 92 observed against 70 expected in IBD and 265 against 235 in GvHD. The two signals are therefore close to independent in all three cohorts, and only mildly enriched in two of them. As can be seen in Figure 4, the CRISPR cells and the glasso cells occupy the grid almost without meeting.

The glasso values themselves are concentrated near zero. In the CRC cohort, W has a mean of 0.083 and a standard deviation of 0.182, 57% of the observed cells are exactly zero and only 5% exceed 0.5 (Figure 5), and the other two cohorts behave the same way. A regression on the continuous W therefore faces a heavily zero-inflated target. For this reason we also treat the thresholded version as a separate binary task.

The CRISPR network is concentrated on a small number of bacteria. Its structure is described here on the full grid rather than on the mask, since the mask block truncates the most connected genera. The median genus carries two or three spacer matches, while the most connected genus carries 54 in CRC, 124 in IBD and 156 in GvHD. Also, in every cohort the ten most connected genera account for about 60% of all CRISPR edges. On the other side, the viruses are far more even, with a median of two matches per vOTU and a maximum between 8 and 12. Therefore, the bipartite network is highly asymmetric: a few bacteria are linked to many phages, but hardly any phage is linked to many bacteria (Figure 15).

Finally, both feature matrices are sparse, and the size asymmetry reported in Section 3.2 is measured there over all profiled entities rather than over the masked pairs (Figures 18 and 16).

How these features are selected, transformed and combined into a representation of a pair is described in Section 4.

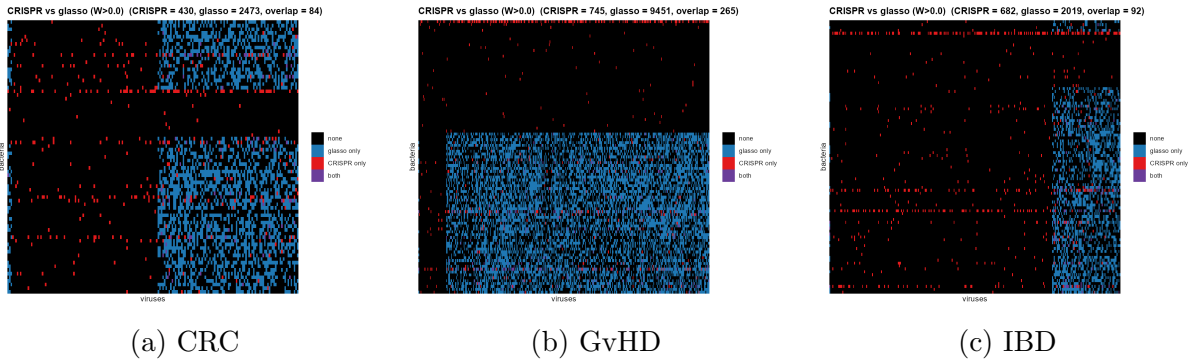


Figure 4: CRISPR and glasso edges on the grid, by cohort: CRISPR only (red), glasso only (blue), both (purple), unobserved (background)

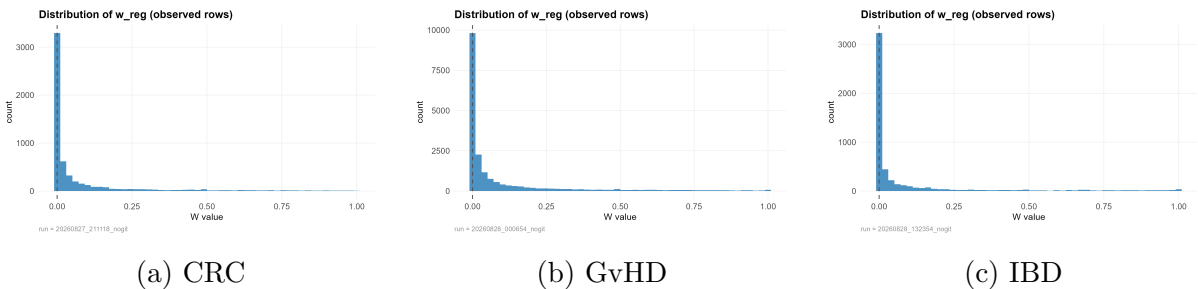


Figure 5: Glasso edge probability on the observed cells, by cohort

4 Methods

4.1 Overview

This chapter describes how the grid of Section 3 is turned into a prediction problem, and what is fitted on it. The same steps are applied to each of the three cohorts separately, since the cohorts do not share a feature vocabulary.

Of the four matrices of Section 3, Y and W supply the prediction targets y , w_{reg} and w_{class} , while X_b and X_v supply the features. Since the observation mask M restricts W to a block of the full grid, we take all three targets on the pairs inside that block. This ensures that each model is fitted and evaluated on the same pairs.

The protein features are not used in raw form. First, we reduce X_b and X_v to a common set of protein clusters, then transform the remaining entries and finally combine the genus profile with the vOTU profile into a single vector describing their pair. Each of the three steps is a choice rather than a necessity, and we return to them for further details in Section 4.3.

Next, we divide the pairs into ten stratified folds. We fit each model on the training set and evaluate on the held-out test set. The models we fit are a sparse logistic regression, a ridge regression, gradient-boosted trees, a single-task neural network for each target, and a joint network that predicts y and w_{class} from one shared representation. Since all of them are fitted on the same folds, they can be compared fold by fold rather than only on average.

Two groups of analyses follow. The first asks whether the results depend on how the features were built, on where the glasso values were thresholded, or on the taxonomic level at which the predictions are read. The second asks what the shared representation of the joint model contributes. How the pipeline was configured and run is described in Section 5.

4.2 Task definitions

The three targets of Section 3 enter the pipeline as three separate prediction problems, in which a pair is described by the same feature vector throughout. Two of them are binary classifications and one a continuous regression. In y we predict whether genus i and vOTU j are linked by a CRISPR spacer, and in w_{class} whether the glasso edge between them was selected in at least one subsample. In w_{reg} we predict the selection frequency W_{ij} directly, so that the model is asked for the value of the interaction and not only for whether it exceeds zero.

Since W_{ij} is bounded between zero and one, nothing prevents the model from predicting a value outside that range. Therefore, we clip the target away from the two endpoints, take its logit, and fit the regression on the result. The predictions are transformed back to the original scale before any metric is computed, so the reported R^2 refers to the original scale of W and not to the logit scale on which the model was fitted.

4.3 Feature representation

4.3.1 Building the pair vector

It is not feasible to pass raw protein-cluster profiles to the model since they are highly zero-inflated and high dimensional. Thus, we reduce the two matrices before joining them since many entries carry zeroes with no information. We do the reduction in three steps. Each step is a choice with two settings: which columns are kept, what the kept entries are turned into, and how the genus profile and the vOTU profile are combined into a single vector. None of the three is a necessity.

Selection ranks the columns of a matrix by their variance and keeps the highest (Figures 17 and 18). The score is always computed on the raw counts and never on a transformed matrix, so the transform of the next step cannot change which clusters are kept and the two steps stay independent of each other. We keep the two sides at equal width: each viral cluster is kept and the bacterial matrix is cut to the same number by taking its k columns of highest variance, with k the viral width. As Section 3 describes, X_b spans an order of magnitude more protein clusters than X_v in every cohort, so the width of the pair vector is set by the viral side and most of the bacterial side is discarded. How much of it survives is therefore not a quantity we set, but a consequence of how many viral clusters a cohort contributes.

We then reduce the kept entries to presence/absence, so that an entry states whether the organism carries at least one protein of that cluster and not how many it carries. The counts are discarded entirely.

Finally, we join the viral and bacterial profiles by adding them position by position, so that the pair vector has the width of one side rather than of both. After the joining we have a vector with coordinates that are either zero, one or two. This is possible only because selection made the widths equal. Bacteria and viruses have separate protein-cluster vocabularies, which means that bacterial cluster j and viral cluster j are unrelated to each other and their sum is not a statement about one specific protein family being present in either partner. It is an architectural choice that halves the width of the pair vector and makes the model share one weight between the two sides.

We did not fix these three settings in advance. Each was compared against its alternative before anything else was fitted, and Section 4.3.2 describes that comparison and the rule by which the settings used here were selected. What is fitted on the resulting pair vectors is described in Section 4.4.

4.3.2 Selecting the representation

Section 4.3.1 describes one representation, but each of its three choices could have been taken differently, and a model reads a different pair vector under every combination of those settings. We call one such combination an arm. Therefore, before committing to a single arm, we first performed an ablation experiment to determine which arm performed best.

The first choice is the preprocessing step that determines which protein features are filtered. Equal-width selection keeps each viral cluster and the same number of bacterial ones that explain the most variance. The alternative keeps a fixed fraction of each of

the two protein-feature matrices, the top 2.5% of X_b and the top 10% of X_v by variance. This alternative leaves the bacterial half of the pair vector far wider than the viral half. The comparison asks whether that imbalance matters, since a model given many more bacterial than viral coordinates may favour the bacterial side for no reason other than its width.

The second choice is what the kept entries become. Presence/absence records only whether a cluster occurs, while the centred log-ratio preserves how large a cluster is relative to the others in the same profile and is the usual treatment of compositional count data. The comparison asks whether the abundance of a protein cluster carries anything beyond its presence.

The third choice is how the profiles are joined. Summation adds them position by position whereas a concatenation leaves each cluster its own coordinate and outputs a pair vector as wide as both sides together. The comparison asks what is lost by folding the two halves onto one another. Since a vector sum requires the two sides to have the same width, this choice is dependent on the first, and equal-width selection has to enter the ladder before summation.

We compare the choices along a ladder of four arms rather than by analysing the full grid and also measuring the interaction between them. The ladder, listed in Table 4, begins from the arm in which each choice takes the setting that was not adopted, and changes exactly one choice at each step until it arrives at the one used in this thesis. Because only one choice moves at a time, the difference between two neighbouring arms corresponds to the effect of that choice. During the ablation we use the same ten folds and the same baseline models as the main experiment. We evaluate and report the ablation on the y and w_{class} tasks.

Table 4: The four ablation arms

Arm	Transform	Selection	Combine
1	centred log-ratio	quantile	concatenation
2	presence/absence	quantile	concatenation
3	presence/absence	equal width	concatenation
4	presence/absence	equal width	summation

4.4 Models

4.4.1 Baselines

As baselines we fit a sparse logistic regression to quantify the linearly learnable structure of the two classification tasks, a ridge regression to quantify it for the regression task, and gradient-boosted trees as well as a neural network to quantify the nonlinearly learnable structure of both. Gradient-boosted trees carry a second role, since they are among the strongest general-purpose predictors on tabular data (Chen and Guestrin, 2016, Grinsztajn et al., 2022), meaning they also serve as a demanding reference against which the neural models are read.

Throughout, $x_i \in \mathbb{R}^d$ is the pair vector of Section 4.3.1, t_i its label on the task at hand and n the number of pairs in the training partition of a fold. We fit every model except the

gradient-boosted trees on the standardised pair vector $\tilde{x}_i$, scale and standardize on the training partition because a tree is unaffected by a monotone rescaling of a coordinate. The sparse logistic regression models the log-odds of the label as a linear function of the pair vector, penalised in L_1 norm,

$$\hat{\beta} = \arg \min_{\beta} \left\{ -C \sum_{i=1}^n c_i [t_i \log p_i + (1 - t_i) \log(1 - p_i)] + \|\beta\|_1 \right\}, \quad p_i = \sigma(\beta_0 + \beta^\top \tilde{x}_i), \quad (4)$$

with $\sigma(z) = (1 + e^{-z})^{-1}$ the logistic function. The L_1 penalty sets most coefficients to exactly zero, so the fitted model rests on a small number of coordinates instead of spreading weight over all of them. The weights c_i are inversely proportional to the frequency of the class of pair i , which keeps the rare positives from being outweighed by the negatives.

For the regression task w_{reg} , we fit a ridge regression, where t_i is the transformed target of Section 4.2. Instead of cross-entropy and an L_1 penalty, we use squared error and an L_2 penalty,

$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^n (t_i - \beta_0 - \beta^\top \tilde{x}_i)^2 + \alpha \|\beta\|_2^2 \right\}. \quad (5)$$

Unlike the L_1 penalty, the L_2 penalty shrinks the coefficients towards zero without setting any of them to it. This means that the fit keeps each of the coefficients and no selection takes place.

The gradient-boosted trees build an additive model out of regression trees, fitted one after another,

$$F_M(x) = F_0 + \nu \sum_{m=1}^M f_m(x), \quad (6)$$

where each tree f_m is fitted to the negative gradient of the loss at the current fit F_{m-1} and scaled by a learning rate ν , with F_0 a constant initial fit and M the number of trees. Since a tree splits on one coordinate after another, the model can represent a response that depends on a combination of coordinates, which a linear function of those same coordinates cannot.

The neural baseline is a network with a single hidden layer,

$$h_i = \max(0, U_1 \tilde{x}_i + b_1) \in \mathbb{R}^H, \quad \eta_i = u_2^\top \delta(h_i) + b_2, \quad p_i = \sigma(\eta_i), \quad (7)$$

with $U_1 \in \mathbb{R}^{H \times d}$ and δ dropping a random share of the hidden units during training and acting as the identity at test time. On the classification tasks we fit the network on a class weighted cross-entropy, in which the positive class is scaled by the inverse of its frequency in the training partition and the negative class is left unweighted. We reweight the gradient-boosted trees in the same way. On the regression task the output is η_i itself and we fit on its squared error. For each task we fit a separate network with the same structure and hyperparameters.

4.4.2 Latent space models

In order to test the claim of the thesis as stated in the introduction chapter, we now introduce latent space models. In a latent space model, we first map the input features to

a latent representation h_i with a neural network and then map that representation to the output with a single linear layer. The latent representation is the h_i of (7), taken before dropout, so that it is exactly the vector the linear layer reads at test time.

During the pipeline we fit three different latent space models. The first two are single task latent space models where one is fitted on y and one on w_{class} , such that each latent space model arranges its latent space for one single task. The third latent space model is a joint model fitted for both targets simultaneously. It keeps everything of (7) before the output layer and replaces that single output by two, such that one latent representation feeds a head for y and a head for w_{class} ,

$$\eta_i^y = u_y^\top \delta(h_i) + b_y, \quad \eta_i^w = u_w^\top \delta(h_i) + b_w, \quad (8)$$

with u_y and u_w the weight vectors of the two heads and b_y and b_w their biases. Writing ℓ for the class-weighted cross-entropy of Section 4.4.1 and t_i^y and t_i^w for the two labels of pair i , we fit the network on the sum of the two losses,

$$\mathcal{L} = \lambda_y \ell(\eta^y, t^y) + \lambda_w \ell(\eta^w, t^w), \quad (9)$$

where we set $\lambda_y = \lambda_w = 1$, so that neither target is given priority over the other. Each head carries its own class weights, since the two targets differ in prevalence.

Apart from the number of heads, the three latent space models are identical in structure and hyperparameters, so a comparison between them isolates the effect of fitting one latent space for two targets rather than for one. The regression task has no joint counterpart, since both heads are classifications.

4.5 Evaluation protocol

We evaluate each model by ten-fold cross-validation over the pairs, stratified on the y label, so that each pair lies in the test partition exactly once. Each model, task and ablation arm of a cohort uses the same ten folds, which means any two models can be compared fold by fold rather than only on their averages. We stratify on y throughout, since one shared set of folds is what makes the tasks comparable and y is the rarer of the two binary targets.

The folds are drawn over pairs and not over organisms, so every genus and every vOTU appears in both partitions and only their combination is new at test time. The evaluation is in this sense transductive and does not describe how a model would behave on an organism it has never seen.

On the binary tasks, we report the AUC score together with the average precision, since at the positive rates of Table 3 the first does not distinguish a good ranking overall from one whose top is mostly wrong. On the regression task, we report R^2 on the original scale of W .

Since every model of a cohort sees the same ten folds, we compare two models by pairing their per-fold test scores and reporting the p -value of a two-sided paired t -test over the ten differences. These p -values are descriptive. Any two of the ten training partitions share most of their pairs, so the differences are not independent, and there is no unbiased estimator of the variance of a cross-validated score (Bengio and Grandvalet, 2004). The variance is therefore underestimated and the test is anti-conservative. We also report

many comparisons without correcting for their number. We do not read a small p -value as evidence against a null hypothesis or as a proper tool of statistical inference, but only as a difference that holds in fold after fold.

4.6 Robustness analyses

Two further analyses ask whether the w_{class} result reflects the target or the way we defined it. The first varies where the continuous W is cut, and the second varies the taxonomic level at which the predictions are read. Either could have produced a target that is easier to predict than the one of Section 4.2, and each holds everything else fixed, so that a difference can be attributed to the choice that moved.

4.6.1 Binarisation threshold

Section 4.2 defines w_{class} by thresholding W at zero, which admits every pair selected in at least one of the fifty subsamples. A stricter cut would define a different target and therefore may change the prediction quality. In order to determine which cut yields a w_{class} with a better prediction quality, we repeat the classification task at ten evenly spaced thresholds inside $(0, 1)$, refitting at each on the label $\mathbf{1}[W_{ij} > t]$. At each threshold, we fit an L_2 -penalised logistic regression ($C = 1.0$) and gradient-boosted trees, so that the answer does not rest on one model. The reason why we fit the regression with an L_2 penalty rather than an L_1 is that the L_1 is too slow on a CPU laptop and starts to become a bottleneck in this task which would make the pipeline infeasible on it.

4.6.2 Taxonomic resolution

Predictions are made at genus level, but a phage infecting one member of a bacterial family may infect others, so the family may be the resolution at which the interaction is better defined. Therefore, we read the same predictions at family level, without refitting any model, and aggregate to family-level predictions following these steps: for each family, we take the median predicted probability over its member genera as the family-level score, and label a family positive when the majority of its member genera carry a positive label. We then aggregate within each fold, such that a family-level score carries a standard deviation across folds. Since a median over several members varies less than a single prediction does, aggregating may raise the measured performance whether or not the family level carries information, which is worth keeping in mind when reading the comparison.

4.7 The shuffled- Y control

Section 4.4.2 fits one latent space for two targets, and if that helps w_{class} there are two reasons it might. Either y carries structure that w_{class} can use, or the second head merely constrains the shared representation and the content of its labels does not matter. We separate the two alternatives by refitting the joint model with the y training labels permuted.

We apply the permutation inside the training partition only. The w_{class} labels stay as they were before, and each label the model is evaluated against stays real, such that

the only thing that changes is what the y head is taught. We use the same folds, the same representation and the same seed as the joint model of Section 4.4.2, and repeat the permutation five times in each fold.

We state the reading in advance. If the real labels lift w_{class} above the permuted ones, then y carries structure that the shared representation passes to w_{class} . If the two are equal, the benefit is the constraint of a second head and the content of its labels is irrelevant. The y head itself is the check that the permutation took effect, since a model taught permuted labels should not predict the real ones.

5 Implementation and reproducibility

5.1 Hyperparameters

We fit all models with the fixed hyperparameters listed in Table 5, identical across tasks, cohorts and ablation arms. We did no hyperparameter fine tuning during the pipeline, since tuning only some arms would confound the comparison of Section 4.

Table 5: Model hyperparameters

Model	Setting
Sparse logistic (<code>sparse_logistic</code>)	L_1 penalty, $C = 1.0$, <code>liblinear</code> solver
XGBoost (<code>xgb_clf</code> , <code>xgb_reg</code>)	500 trees, max depth 4, learning rate 0.05, subsample and column subsample 0.8
Ridge regression (<code>linreg</code>)	$\alpha = 1.0$
MLP, single-task (<code>mlp_clf_y</code> , <code>mlp_clf_w_class</code> , <code>mlp_reg</code> , <code>mlp_latent_y</code> , <code>mlp_latent_w_class</code>)	hidden width 64, dropout 0.3, batch size 512, learning rate 3×10^{-4} , weight decay 10^{-3} , 200 epochs
MLP, joint (<code>mlp_latent_yw</code>)	as above, plus loss weights $\lambda_y = \lambda_w = 1.0$

5.2 Implementation and reproducibility

The full pipeline, from raw data to the figures and tables in this thesis, is implemented in code provided in electronic form (Section B), together with the package versions and installation instructions needed to run it.

Every stage runs under a fixed random seed, so that the ten folds, model initialisations and any random subsampling are identical across repeated runs of the same configuration. This fixes the stochastic parts of the pipeline but does not guarantee bit-identical results across different hardware or library versions.

6 Results

6.1 Representation ablation

Table 4 defines the four-arm ladder, evaluated on the same folds and models as the main experiment, on y and w_{class} ; per-arm means and standard deviations for every cohort are given in Table 13. We fix the arm throughout the pipeline with a rule: an arm is adopted over its predecessor only if the mean AUC gap exceeds one fold’s standard deviation.

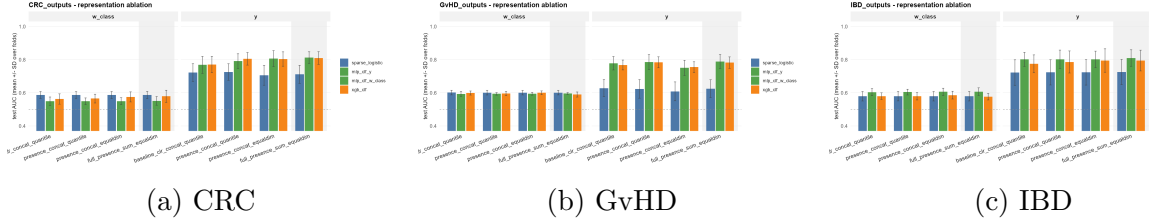


Figure 6: Representation ablation: test AUC by arm, model and task

On the y task, arm 4 gives the highest, or near highest, mean AUC for both nonlinear models in each cohort (CRC: MLP 0.812, XGBoost 0.810; GvHD: MLP 0.789, XGBoost 0.782; IBD: MLP 0.810, XGBoost 0.794; Figure 6). In CRC, this comes from a single step: swapping the CLR-transformation for presence/absence (arm 1 $\rightarrow$ 2) gives an incremental advantage to the nonlinear models (MLP +0.023 against a fold-gap SD of 0.012; XGBoost +0.034 against 0.025) but not to sparse logistic (+0.003 against 0.009), and adding equal-dimension selection and summation (arm 2 $\rightarrow$ 4) moves both models by less than their own fold-gap SD, so neither factor is decisive on its own. In GvHD, presence/absence only clears the bar for XGBoost (+0.016 against 0.015); equal-dimension selection then drops both nonlinear models and sparse logistic (MLP -0.036 against 0.033; XGBoost -0.029 against 0.014; sparse logistic -0.015 against 0.010), and summation recovers almost the same amount (MLP +0.039 against 0.031; XGBoost +0.027 against 0.012; sparse logistic +0.016 against 0.008), so arm 1 $\rightarrow$ 4 is not decisive overall. In IBD, no step clears the bar for any model. Sparse logistic only moves opposite to the nonlinear models in CRC, over the full ladder (-0.011 against 0.010), driven mainly by equal-dimension selection (-0.020 against 0.015).

For CRC and IBD, on w_{class} , no arm is decisive. For GvHD however there are some decisive factors. For example for the XGBoost model, arm 4 underperforms arm 1 (-0.009 against 0.006). Besides XGBoost, no other model is decisive either. Because of this reason, we cannot choose an arm solely based on the w_{class} task result and need to rely on the y task results. Additionally, we select the arm based on the MLP alone, since that is the model the main experiment is built on via the latent space (Section 4.4.2). XGBoost’s agreement corroborates the choice, without being decisive on its own. We keep the sparse logistic regression fixed as an unoptimised linear reference.

6.2 Binarisation threshold

As will be discussed in Section 6.4.2, w_{reg} performs poorly across all cohorts, which is why we binarise it into w_{class} . As explained in Section 4.6.1, we do this by fitting two different

models across ten different cutoffs, in order to determine which one is the most optimal.

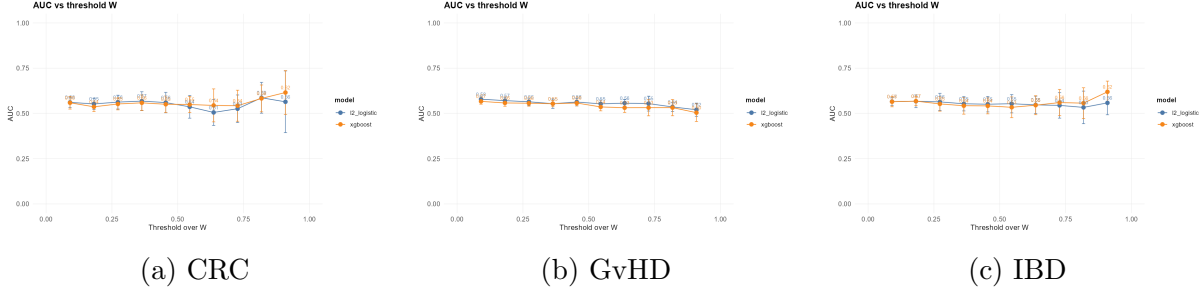


Figure 7: Test AUC against the binarisation threshold t on W

No cohort or model exceeds 0.62 AUC at any interior threshold (Figure 7). Additionally, the variance of the AUC grows as the threshold rises. The reason is that with an increasing threshold, the number of pairs that belong to the positive class shrinks: at $t = 0$, w_{class} is close to balanced (43% positive in CRC, 49% in GvHD, 38% in IBD) but by $t \approx 0.5$ only 5–7% of pairs remain positive, and the sweep excludes $t = 1$ itself, where $\mathbf{1}[W > t]$ would contain no positive pairs at all. That’s consistent with $t = 0$ being the best cutoff among those tested for the w_{class} task.

6.3 Taxonomic resolution

Genus-level AUC is slightly higher than family-level AUC across all three cohorts, on both tasks, with only two trivial exceptions in GvHD (Figure 8; the underlying numbers are in Table 14). Most families are singletons (18 of 27 in CRC, 15 of 32 in GvHD, 20 of 34 in IBD), where genus and family level coincide by construction and share the same label. The remaining, multi-member families do not always share a consistent label across their genera, so the median aggregation used to collapse them discards real heterogeneity, which causes the slight drop in AUC. On y , where this effect is real, sparse logistic loses the most from it, since its per-genus probabilities carry the most direct signal, while the nonlinear models lose comparatively little. On w_{class} , where no real effect exists to begin with, no consistent pattern emerges across cohorts or models. We therefore base the resolution choice on y alone and fix genus level throughout the rest of the pipeline.

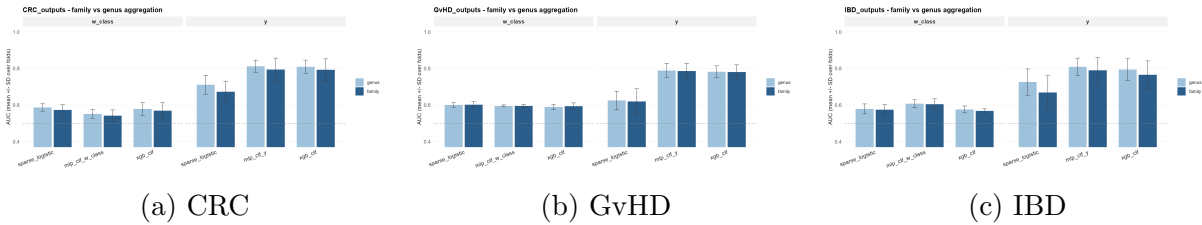


Figure 8: Genus- vs family-level AUC, by task and model

6.4 Prediction performance

We report AUC and average precision (AP) for y , w_{class} , and w_{reg} across all three cohorts, one subsubsection per task, with baseline and latent-space models reported together.

6.4.1 y

Table 6: Test performance on y

Cohort	Model	AUC	AP
CRC	Sparse logistic	0.711	0.165
	XGBoost	0.810	0.259
	MLP (baseline)	0.812	0.263
	MLP (latent, single-task)	0.812	0.263
	MLP (joint)	0.810	0.276
IBD	Sparse logistic	0.726	0.243
	XGBoost	0.794	0.283
	MLP (baseline)	0.810	0.285
	MLP (latent, single-task)	0.810	0.285
	MLP (joint)	0.802	0.271
GvHD	Sparse logistic	0.625	0.074
	XGBoost	0.782	0.151
	MLP (baseline)	0.789	0.205
	MLP (latent, single-task)	0.789	0.205
	MLP (joint)	0.789	0.191

The baseline MLP beats sparse logistic regression on AUC in each fold of every cohort (CRC $+0.101$, $p = 6 \times 10^{-5}$; IBD $+0.084$, $p = 4 \times 10^{-4}$; GvHD $+0.164$, $p < 10^{-4}$; Table 6), indicating that the y signal in the pair representation is not linearly separable. The individual folds behind Table 6 are listed in Table 10 and drawn in Figures 10 and 11. AP reinforces the outcome as well. The AP gap is significant in CRC ($p = 4 \times 10^{-4}$) and GvHD ($p = 2 \times 10^{-5}$) but not in IBD ($p = 0.078$), where sparse logistic’s ranking of the top candidates is closer to the nonlinear models’ than its AUC alone suggests.

Against XGBoost, the MLP shows no AUC difference in any cohort (CRC $p = 0.80$; IBD $p = 0.13$; GvHD $p = 0.52$). However, in GvHD, the MLP beats XGBoost on AP ($+0.054$, $p = 0.027$), meaning the two models rank the full set equally well but the MLP places the highest-confidence pairs more accurately.

The joint model leaves y AUC unchanged in every cohort (CRC $p = 0.55$; IBD $p = 0.25$; GvHD $p = 0.99$). However, AP in IBD shows a small significant drop (-0.014 , $p = 0.022$), while CRC and GvHD show no difference ($p = 0.40$, $p = 0.47$). Therefore, the second head added for w_{class} does not increase y task prediction, and in fact hurts it slightly in IBD.

6.4.2 w_{reg}

 Table 7: Test R^2 on w_{reg}

Cohort	Model	R^2
CRC	Ridge regression	−0.156
	XGBoost	−0.154
	MLP	−0.141
IBD	Ridge regression	−0.153
	XGBoost	−0.145
	MLP	−0.123
GvHD	Ridge regression	−0.172
	XGBoost	−0.180
	MLP	−0.144

R^2 is negative for every model in every cohort (Table 7), meaning each model performs worse than predicting the mean, and no individual fold of any model reaches zero (Table 12, Figure 14). This is why we binarise W into w_{class} , evaluated next.

6.4.3 w_{class}

 Table 8: Test performance on w_{class} ; the latent single-task MLP equals the baseline MLP in every fold

Cohort	Model	AUC	AP
CRC	Sparse logistic	0.587	0.502
	XGBoost	0.579	0.492
	MLP (baseline)	0.551	0.473
	MLP (latent, single-task)	0.551	0.473
	MLP (joint)	0.575	0.496
IBD	Sparse logistic	0.579	0.450
	XGBoost	0.576	0.450
	MLP (baseline)	0.607	0.478
	MLP (latent, single-task)	0.607	0.478
	MLP (joint)	0.608	0.475
GvHD	Sparse logistic	0.601	0.587
	XGBoost	0.590	0.572
	MLP (baseline)	0.596	0.577
	MLP (latent, single-task)	0.596	0.577
	MLP (joint)	0.601	0.587

Every model in every cohort sits between 0.551 and 0.608 AUC (Table 8; per-fold values in Table 11, distributions in Figures 12 and 13). That range sits far lower than y ’s. Unlike y , added model capacity does not consistently help: the baseline MLP is significantly *worse* than sparse logistic in CRC on both AUC (−0.036, $p = 4 \times 10^{-4}$) and AP (−0.029,

$p = 1 \times 10^{-4}$), significantly *better* in IBD (AUC +0.028, $p = 0.013$; AP +0.028, $p = 0.023$), and not clearly different in GvHD (AUC -0.005 , $p = 0.093$; AP -0.009 , $p = 0.018$). XGBoost never beats sparse logistic: it ties in CRC and IBD ($p \geq 0.28$ on both metrics) and loses significantly in GvHD (AUC $p = 0.005$; AP $p < 10^{-4}$).

The joint model outperforms the single-task MLP in two of the three cohorts: clearly in CRC (+0.023 AUC, $p = 0.002$) and more narrowly in GvHD (+0.006, $p = 0.049$), with no difference in IBD (+0.001, $p = 0.876$). Against logistic regression, the joint model loses in CRC (-0.012 , $p = 0.005$) while beating it clearly in IBD (+0.029, $p = 0.002$) and matching it in GvHD ($p = 0.911$). Whether the CRC and GvHD gains over the single-task MLP reflect structure genuinely shared with y , or only the regularising effect of training a second head, is tested and discussed in the next subsection.

6.5 Joint training and the shuffled- Y control

To test whether the joint model’s benefit to w_{class} (Section 6.4.3) reflects structure genuinely shared with y , or only the regularising effect of fitting a second head, we apply the control of Section 4.7: we retrain the joint MLP with the y labels permuted within each outer-fold training set, repeat this over five independent permutations, and evaluate both heads on the unpermuted test fold. If the w_{class} gain survives even when y carries no real signal, that gain cannot come from y itself.

Table 9: Shuffled- y control: joint model performance, real vs permuted y labels (differences: real – permuted)

Cohort	Head	y labels	AUC	AP
CRC	y	Real	0.810	0.276
		Permuted	0.485	0.042
	w_{class}	Real	0.575	0.496
		Permuted	0.579	0.497
IBD	y	Real	0.802	0.271
		Permuted	0.495	0.042
	w_{class}	Real	0.608	0.475
		Permuted	0.604	0.472
GvHD	y	Real	0.789	0.191
		Permuted	0.484	0.026
	w_{class}	Real	0.601	0.587
		Permuted	0.602	0.587

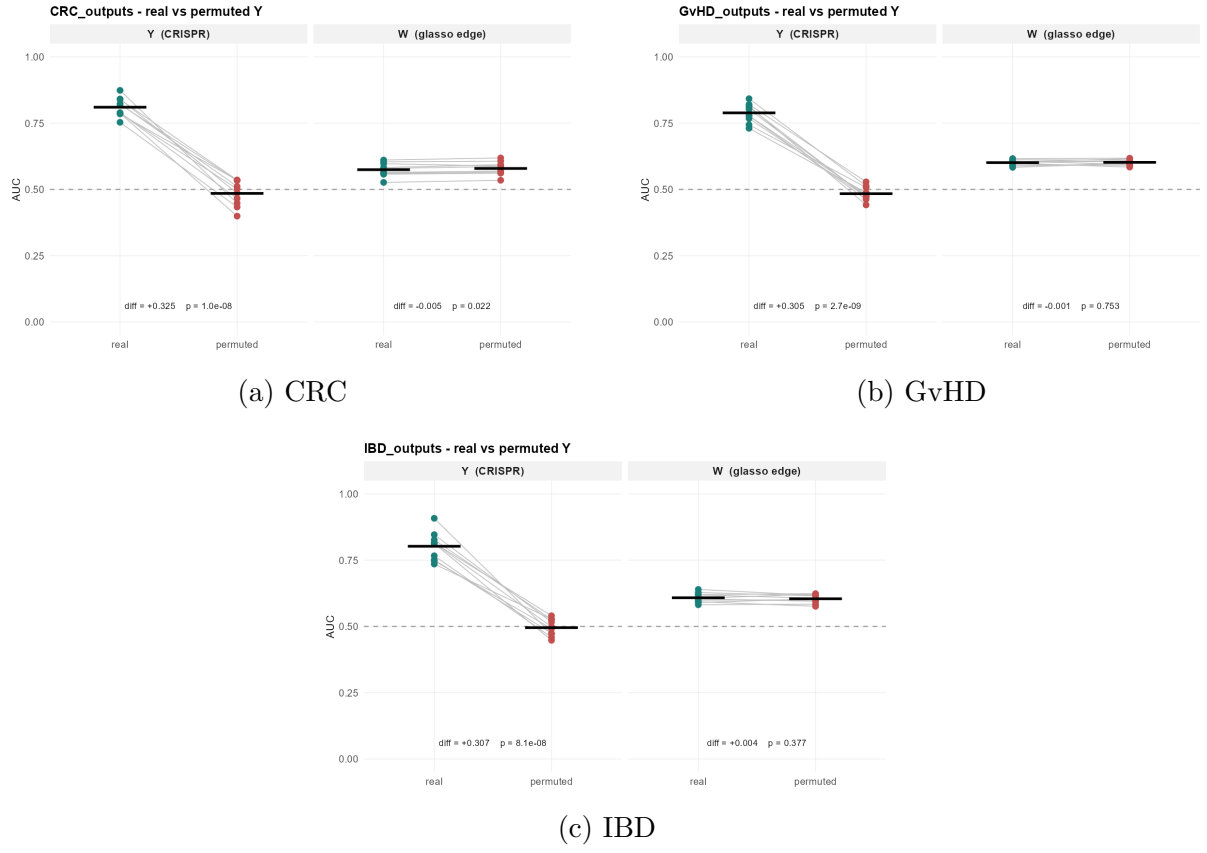


Figure 9: Real versus permuted y labels, per outer fold. Each grey line joins one fold's real and permuted value, the black bar is the mean and the dashed line is chance. Only the y training labels are permuted; the w_{class} labels and every evaluation label stay real.

The permutation removes the intended signal from the y head: performance collapses to chance in every cohort (AUC 0.484–0.495 against 0.789–0.810 with the real labels; AP 0.026–0.042 against 0.191–0.276; Table 9), and the drop is significant throughout ($p < 10^{-7}$ on AUC and $p \leq 2 \times 10^{-4}$ on AP, all cohorts).

In the w_{class} task, permuting y leaves its performance unchanged in IBD (AUC +0.004, $p = 0.377$; AP +0.003, $p = 0.228$) and GvHD (AUC −0.001, $p = 0.753$; AP +0.000, $p = 0.994$). On the other hand, in the CRC cohort, it produces a small but significant change in the opposite direction (AUC −0.005, $p = 0.022$; AP not significant, $p = 0.828$). With real labels the model does very slightly *worse* on w_{class} than with permuted ones. In no cohort does permuting y significantly reduce w_{class} performance. The advantage the joint model holds over the single-task MLP in CRC and GvHD (Section 6.4.3) is reproduced in full by the permuted model, which reaches 0.579 against 0.551 in CRC and 0.602 against 0.596 in GvHD; in IBD there was no advantage to account for. Since that advantage survives destroying y ’s real content entirely, it is not explained by structure shared between y and w_{class} ; it is consistent with the plain regularising effect of fitting a second head, regardless of what that head predicts.

7 Discussion

7.1 Interpretation

The weak signal on the W task barely responds to any of the remedies tried: added capacity helps inconsistently (Section 6.4.3), no arm improves w_{class} in any cohort (Section 6.1), no interior cut of W exceeds 0.62 AUC (Figure 7), and family-level aggregation does not help (Section 6.3). This could be because of a structural mismatch. Y_{ij} records a physical event between two genomes and is for that reason a property of the pair alone, so the biology that defines it also makes it representable from x_i and x_j . W_{ij} has no such definition: it is an entry of a regularised precision matrix and depends on every other organism in the community, so no function of x_i and x_j represents it exactly. This is a conjecture rather than a tested claim, and the consistent above-chance signal makes it a bound rather than a barrier. A second alternative is that W does not measure interaction at all, supported by the targets overlapping little more than independence predicts (Section 3) and by abundance correlations recovering known infection networks at close to chance level (Coenen and Weitz, 2018), a limitation reported for microbial association networks more generally (Pinto et al., 2022).

7.2 Limitations and future work

Four limitations bound these results. The folds are drawn over pairs rather than organisms (Section 4.5), so every genus and vOTU appears in training and only the combination is new at test time; what is measured is the prediction of unobserved pairings among known organisms, not host prediction for an unseen phage. CRISPR linkage is a lower bound on interaction, since a missing spacer does not establish that no infection occurred (Zhang et al., 2021), so the negative class is unreliable. Predictions are made at genus and vOTU level, while host range is frequently strain-specific. And every model uses one fixed hyperparameter setting (Section 5), chosen to keep the arms comparable rather than to maximise performance.

Two directions follow from the representation. Existing approaches build their features from curated resources (Edwards et al., 2016, Coutinho et al., 2021, Roux et al., 2023, Nie et al., 2024), whereas the one used here is annotation-free. If a biological prior is what separates the two targets, then for W the place to supply it is the features, since the target itself cannot be redefined: selecting protein clusters by biological role, such as receptor binding, adhesion or defence, would aim the model at the component of W that a genuine infection would produce rather than at whatever varies most. A protein language model such as ESM-2 (Lin et al., 2023) would supply the same prior in transferred form, and its embeddings are shared across cohorts, which would in addition remove the vocabulary barrier that currently prevents training in one cohort and evaluating in another (Section 3). How far either can go on W is bounded by the mismatch of Section 7.1, since the part of W_{ij} that depends on the rest of the community remains beyond the reach of any pairwise representation.

8 Conclusion

After these experiments, we can conclude that no shared latent space unifying y and w_{class} in terms of prediction has emerged. The joint model matches the single-task network on y , and its small gain on w_{class} survives permuting the y labels (Table 9), so what it learns is a latent space for y with a second head attached rather than a representation serving both.

Beyond the two targets, the pattern suggests something about the approach. The pipeline uses no functional annotation: the protein clusters are formed by sequence similarity without functional labelling, the features are chosen by variance, and the models are generic. The single place where explicit biological knowledge enters is the Y label, which records a mechanism rather than an association, and that is the single place the pipeline performs well. W has no comparable biological prior. It is the output of an estimator applied to abundances (Friedman et al., 2008, Liu et al., 2010), and nothing in the features or the objective constrains it towards infection rather than towards any other reason two organisms might covary.

A Appendix

A.1 Additional figures

Figures supporting statements made in Sections 3 and 6.

A.1.1 Per-fold prediction performance

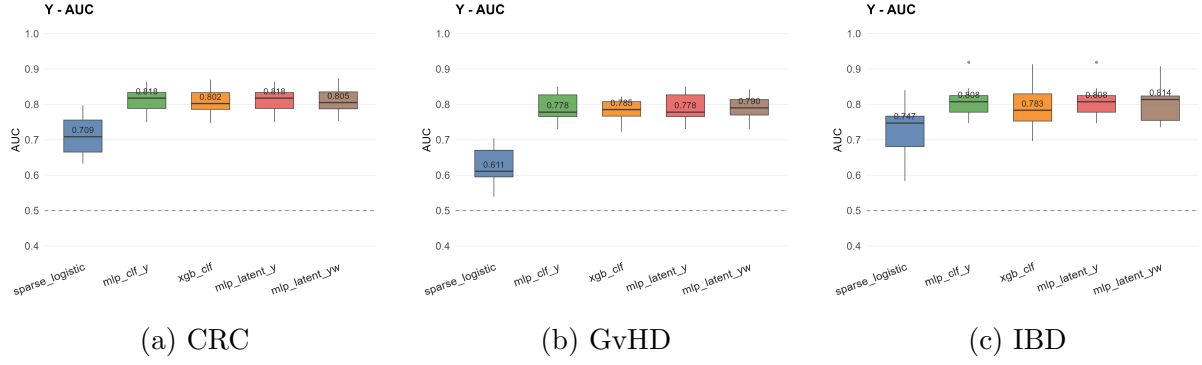


Figure 10: Per-fold test AUC on y

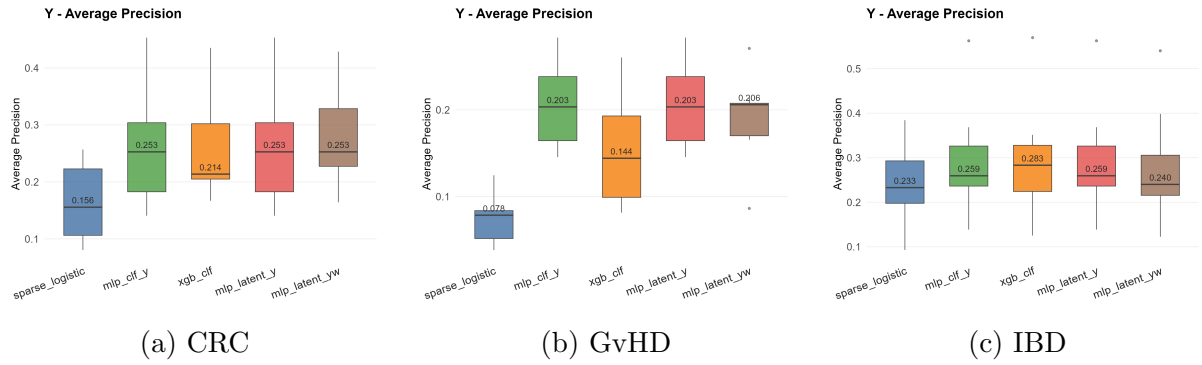
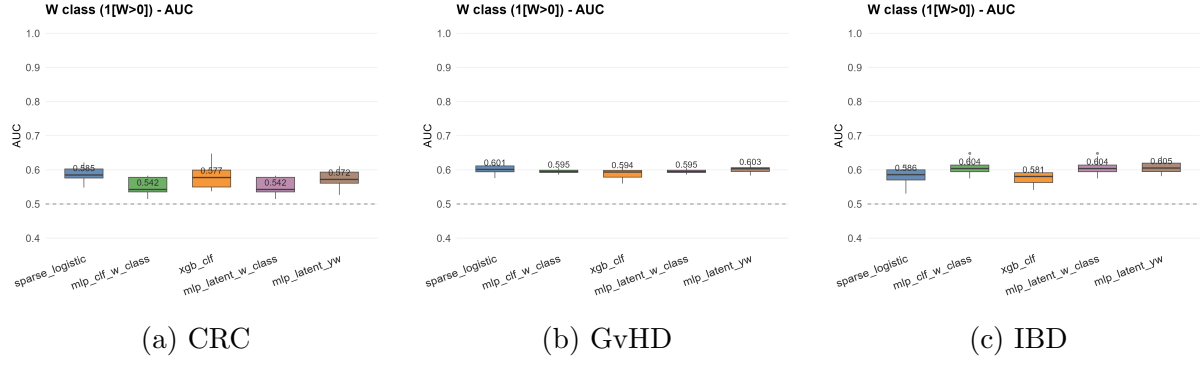
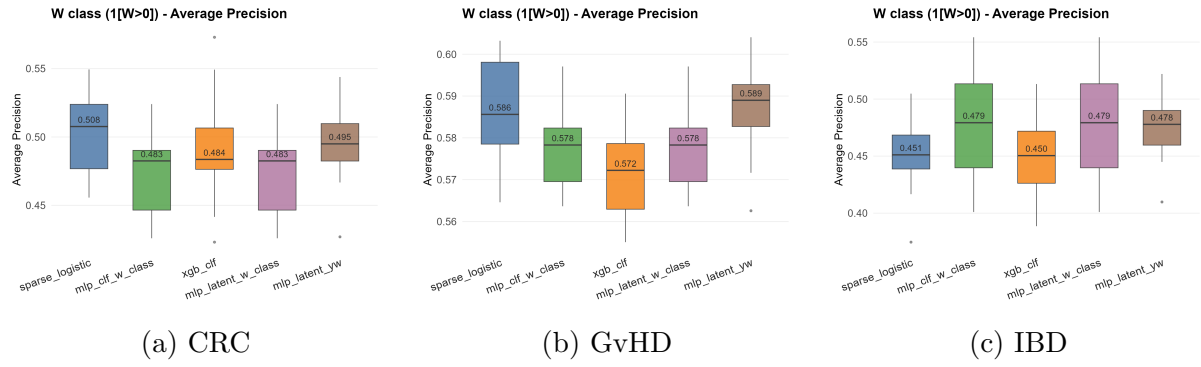
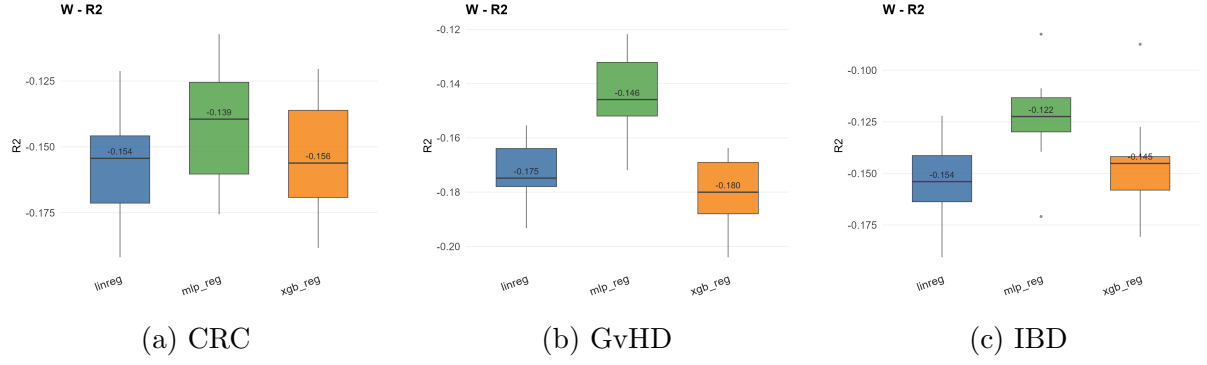


Figure 11: Per-fold test AP on y


 Figure 12: Per-fold test AUC on w_{class}

 Figure 13: Per-fold test AP on w_{class}


 Figure 14: Per-fold test R^2 on w_{reg}

A.1.2 CRISPR network and feature matrices

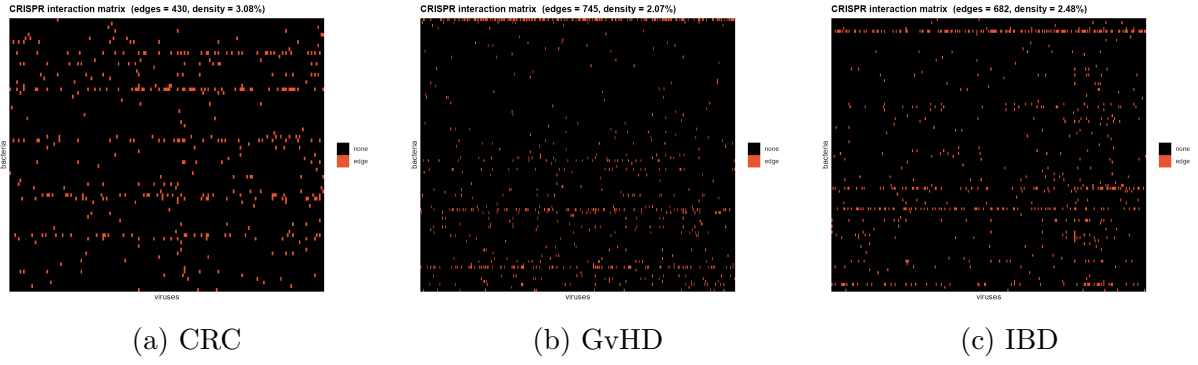
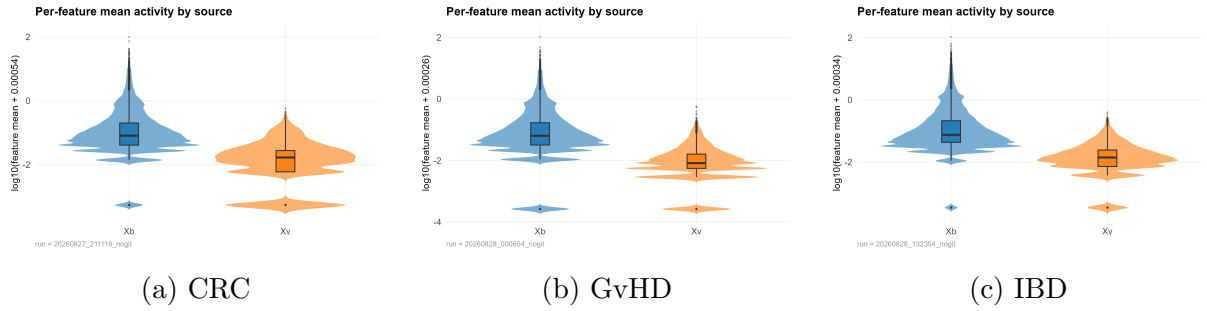

 Figure 15: CRISPR linkage on the full genus $\times$ vOTU grid


Figure 16: Per-feature mean activity by source (log scale)

A.1.3 Variance-based feature selection

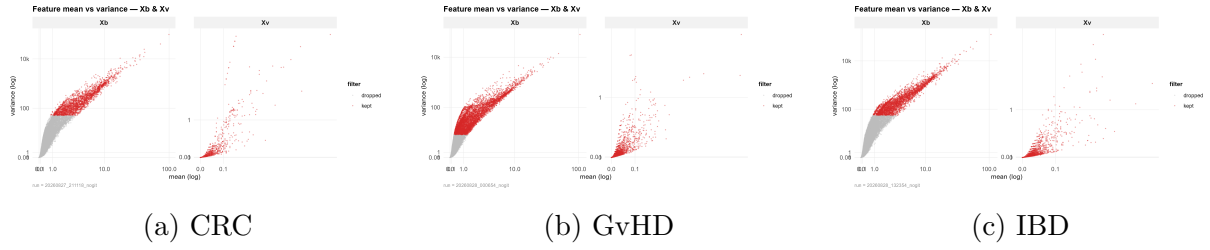


Figure 17: Feature mean vs variance, coloured by variance selection

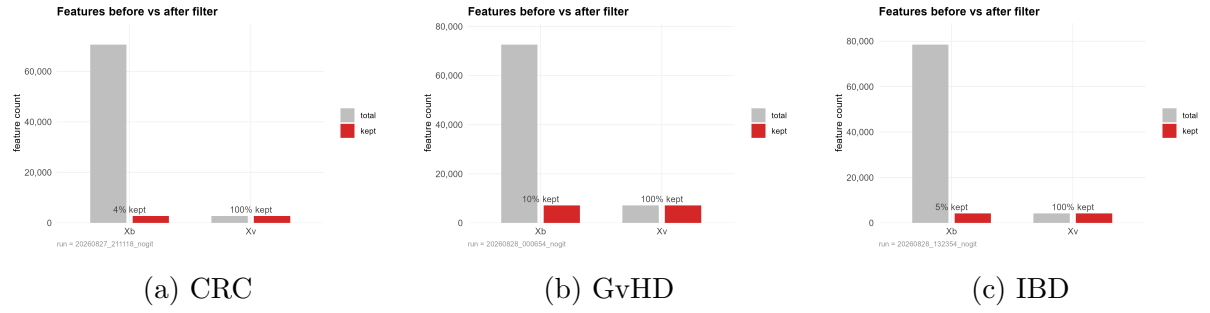


Figure 18: Protein clusters before and after the variance filter

A.2 Full per-fold results

The individual folds behind the means reported in Section 6, on the test partition.

Table 10: Per-fold test performance on y : AUC (AP)

Cohort	Fold	Sparse logistic	XGBoost	MLP (base)	MLP (latent)	MLP (joint)
CRC	0	0.660 (0.129)	0.748 (0.200)	0.750 (0.192)	0.750 (0.192)	0.753 (0.212)
	1	0.673 (0.080)	0.812 (0.167)	0.785 (0.176)	0.785 (0.176)	0.790 (0.225)
	2	0.766 (0.189)	0.870 (0.309)	0.864 (0.277)	0.864 (0.277)	0.874 (0.337)
	3	0.698 (0.254)	0.784 (0.435)	0.800 (0.453)	0.800 (0.453)	0.785 (0.429)
	4	0.797 (0.234)	0.861 (0.359)	0.850 (0.287)	0.850 (0.287)	0.841 (0.333)
	5	0.720 (0.257)	0.786 (0.203)	0.836 (0.384)	0.836 (0.384)	0.840 (0.316)
	6	0.663 (0.099)	0.793 (0.211)	0.824 (0.180)	0.824 (0.180)	0.823 (0.245)
	7	0.633 (0.092)	0.787 (0.216)	0.812 (0.229)	0.812 (0.229)	0.787 (0.235)
	8	0.760 (0.167)	0.820 (0.282)	0.826 (0.310)	0.826 (0.310)	0.821 (0.261)
	9	0.743 (0.144)	0.838 (0.211)	0.777 (0.141)	0.777 (0.141)	0.788 (0.164)
IBD	0	0.681 (0.093)	0.785 (0.125)	0.772 (0.148)	0.772 (0.148)	0.751 (0.136)
	1	0.649 (0.208)	0.749 (0.302)	0.809 (0.260)	0.809 (0.260)	0.815 (0.244)
	2	0.682 (0.194)	0.747 (0.297)	0.747 (0.301)	0.747 (0.301)	0.748 (0.278)
	3	0.583 (0.141)	0.696 (0.125)	0.756 (0.139)	0.756 (0.139)	0.735 (0.122)
	4	0.746 (0.303)	0.782 (0.270)	0.806 (0.244)	0.806 (0.244)	0.767 (0.232)
	5	0.763 (0.375)	0.765 (0.351)	0.796 (0.369)	0.796 (0.369)	0.815 (0.398)
	6	0.840 (0.245)	0.914 (0.222)	0.919 (0.234)	0.919 (0.234)	0.908 (0.210)
	7	0.795 (0.384)	0.858 (0.570)	0.826 (0.563)	0.826 (0.563)	0.846 (0.540)
	8	0.749 (0.221)	0.836 (0.231)	0.846 (0.259)	0.846 (0.259)	0.827 (0.237)
	9	0.768 (0.262)	0.810 (0.337)	0.821 (0.334)	0.821 (0.334)	0.813 (0.315)
GvHD	0	0.620 (0.084)	0.762 (0.215)	0.740 (0.198)	0.740 (0.198)	0.731 (0.166)
	1	0.693 (0.076)	0.823 (0.169)	0.836 (0.238)	0.836 (0.238)	0.842 (0.206)
	2	0.602 (0.062)	0.819 (0.201)	0.811 (0.283)	0.811 (0.283)	0.811 (0.271)
	3	0.601 (0.092)	0.811 (0.163)	0.775 (0.176)	0.775 (0.176)	0.777 (0.208)
	4	0.704 (0.083)	0.781 (0.092)	0.850 (0.252)	0.850 (0.252)	0.815 (0.086)
	5	0.684 (0.125)	0.799 (0.260)	0.781 (0.209)	0.781 (0.209)	0.803 (0.206)
	6	0.630 (0.048)	0.782 (0.107)	0.769 (0.238)	0.769 (0.238)	0.778 (0.207)
	7	0.581 (0.038)	0.789 (0.081)	0.832 (0.151)	0.832 (0.151)	0.821 (0.168)
	8	0.539 (0.047)	0.723 (0.096)	0.765 (0.161)	0.765 (0.161)	0.768 (0.214)
	9	0.593 (0.081)	0.729 (0.126)	0.730 (0.145)	0.730 (0.145)	0.744 (0.176)

Table 11: Per-fold test performance on w_{class} : AUC (AP)

Cohort	Fold	Sparse logistic	XGBoost	MLP (base)	MLP (latent)	MLP (joint)
CRC	0	0.622 (0.530)	0.648 (0.573)	0.582 (0.491)	0.582 (0.491)	0.611 (0.544)
	1	0.581 (0.512)	0.560 (0.479)	0.535 (0.479)	0.535 (0.479)	0.583 (0.508)
	2	0.577 (0.507)	0.540 (0.477)	0.537 (0.487)	0.537 (0.487)	0.557 (0.498)
	3	0.603 (0.509)	0.612 (0.499)	0.583 (0.469)	0.583 (0.469)	0.604 (0.492)
	4	0.569 (0.483)	0.595 (0.509)	0.580 (0.491)	0.580 (0.491)	0.563 (0.491)
	5	0.548 (0.456)	0.537 (0.423)	0.515 (0.426)	0.515 (0.426)	0.527 (0.427)
	6	0.603 (0.549)	0.597 (0.549)	0.574 (0.524)	0.574 (0.524)	0.597 (0.541)
	7	0.589 (0.475)	0.550 (0.442)	0.524 (0.439)	0.524 (0.439)	0.560 (0.467)
	8	0.603 (0.470)	0.600 (0.476)	0.548 (0.434)	0.548 (0.434)	0.579 (0.479)
IBD	9	0.576 (0.528)	0.550 (0.488)	0.535 (0.486)	0.535 (0.486)	0.564 (0.511)
	0	0.530 (0.375)	0.541 (0.389)	0.575 (0.401)	0.575 (0.401)	0.593 (0.410)
	1	0.614 (0.472)	0.600 (0.478)	0.596 (0.459)	0.596 (0.459)	0.628 (0.480)
	2	0.602 (0.487)	0.565 (0.449)	0.640 (0.527)	0.640 (0.527)	0.620 (0.516)
	3	0.532 (0.447)	0.576 (0.483)	0.601 (0.514)	0.601 (0.514)	0.582 (0.492)
	4	0.578 (0.505)	0.585 (0.513)	0.616 (0.554)	0.616 (0.554)	0.599 (0.522)
	5	0.595 (0.447)	0.557 (0.419)	0.584 (0.421)	0.584 (0.421)	0.590 (0.445)
	6	0.590 (0.436)	0.593 (0.454)	0.648 (0.513)	0.648 (0.513)	0.606 (0.476)
	7	0.568 (0.417)	0.562 (0.415)	0.594 (0.433)	0.594 (0.433)	0.604 (0.455)
GvHD	8	0.582 (0.455)	0.586 (0.451)	0.609 (0.491)	0.609 (0.491)	0.640 (0.483)
	9	0.602 (0.458)	0.598 (0.449)	0.607 (0.467)	0.607 (0.467)	0.619 (0.472)
	0	0.576 (0.582)	0.560 (0.560)	0.586 (0.582)	0.586 (0.582)	0.594 (0.590)
	1	0.616 (0.601)	0.597 (0.583)	0.605 (0.575)	0.605 (0.575)	0.604 (0.593)
	2	0.587 (0.565)	0.585 (0.555)	0.594 (0.564)	0.594 (0.564)	0.599 (0.572)
	3	0.613 (0.590)	0.594 (0.573)	0.598 (0.569)	0.598 (0.569)	0.612 (0.593)
	4	0.618 (0.603)	0.607 (0.591)	0.602 (0.584)	0.602 (0.584)	0.617 (0.602)
	5	0.596 (0.576)	0.576 (0.560)	0.593 (0.582)	0.593 (0.582)	0.583 (0.563)
	6	0.607 (0.579)	0.609 (0.571)	0.596 (0.570)	0.596 (0.570)	0.603 (0.582)
GvHD	7	0.595 (0.589)	0.576 (0.571)	0.590 (0.582)	0.590 (0.582)	0.589 (0.588)
	8	0.594 (0.578)	0.599 (0.575)	0.592 (0.567)	0.592 (0.567)	0.607 (0.585)
	9	0.607 (0.602)	0.593 (0.580)	0.599 (0.597)	0.599 (0.597)	0.605 (0.604)

Table 12: Per-fold test R^2 on w_{reg}

Cohort	Fold	Ridge	XGBoost	MLP
CRC	0	-0.147	-0.136	-0.132
	1	-0.172	-0.171	-0.157
	2	-0.170	-0.164	-0.161
	3	-0.121	-0.120	-0.107
	4	-0.145	-0.137	-0.114
	5	-0.156	-0.157	-0.147
	6	-0.180	-0.177	-0.164
	7	-0.128	-0.131	-0.124
	8	-0.153	-0.156	-0.131
	9	-0.192	-0.188	-0.176
IBD	0	-0.166	-0.161	-0.140
	1	-0.127	-0.141	-0.116
	2	-0.172	-0.146	-0.130
	3	-0.158	-0.144	-0.127
	4	-0.191	-0.173	-0.171
	5	-0.122	-0.087	-0.083
	6	-0.156	-0.144	-0.109
	7	-0.142	-0.127	-0.112
	8	-0.152	-0.149	-0.129
	9	-0.141	-0.181	-0.118
GvHD	0	-0.193	-0.204	-0.172
	1	-0.173	-0.183	-0.146
	2	-0.178	-0.177	-0.146
	3	-0.159	-0.164	-0.123
	4	-0.177	-0.185	-0.145
	5	-0.181	-0.189	-0.157
	6	-0.163	-0.168	-0.145
	7	-0.166	-0.172	-0.128
	8	-0.176	-0.191	-0.154
	9	-0.155	-0.166	-0.122

Table 13: Ablation: test AUC, mean (SD) over folds

Cohort	Task	Arm	Sparse logistic	XGBoost	MLP
CRC	y	1	0.722 (0.053)	0.771 (0.049)	0.768 (0.051)
		2	0.725 (0.050)	0.805 (0.038)	0.791 (0.044)
		3	0.706 (0.060)	0.804 (0.043)	0.806 (0.048)
		4	0.711 (0.055)	0.810 (0.038)	0.812 (0.035)
	w_{class}	1	0.587 (0.021)	0.563 (0.031)	0.549 (0.025)
		2	0.587 (0.020)	0.565 (0.027)	0.549 (0.020)
		3	0.587 (0.021)	0.575 (0.030)	0.550 (0.020)
		4	0.587 (0.021)	0.579 (0.037)	0.551 (0.026)
IBD	y	1	0.722 (0.077)	0.775 (0.053)	0.802 (0.042)
		2	0.723 (0.076)	0.786 (0.067)	0.803 (0.055)
		3	0.723 (0.077)	0.795 (0.071)	0.801 (0.050)
		4	0.726 (0.076)	0.794 (0.063)	0.810 (0.050)
	w_{class}	1	0.579 (0.029)	0.580 (0.020)	0.604 (0.021)
		2	0.579 (0.028)	0.579 (0.022)	0.604 (0.018)
		3	0.579 (0.029)	0.586 (0.024)	0.606 (0.022)
		4	0.579 (0.029)	0.576 (0.020)	0.607 (0.023)
GvHD	y	1	0.628 (0.051)	0.768 (0.030)	0.778 (0.041)
		2	0.624 (0.056)	0.784 (0.033)	0.786 (0.044)
		3	0.609 (0.057)	0.754 (0.032)	0.750 (0.046)
		4	0.625 (0.054)	0.782 (0.035)	0.789 (0.042)
	w_{class}	1	0.601 (0.014)	0.599 (0.012)	0.594 (0.014)
		2	0.601 (0.014)	0.595 (0.013)	0.594 (0.008)
		3	0.601 (0.014)	0.600 (0.011)	0.594 (0.008)
		4	0.601 (0.014)	0.590 (0.015)	0.596 (0.006)

Table 14: Genus vs family test AUC, mean (SD) over folds

Cohort	Task	Model	Genus	Family	Difference
CRC	y	Sparse logistic	0.711 (0.052)	0.673 (0.058)	-0.038
		XGBoost	0.810 (0.036)	0.793 (0.060)	-0.017
		MLP (base)	0.812 (0.033)	0.794 (0.063)	-0.018
	w_{class}	Sparse logistic	0.587 (0.020)	0.573 (0.029)	-0.013
		XGBoost	0.579 (0.035)	0.569 (0.043)	-0.009
		MLP (base)	0.551 (0.025)	0.543 (0.031)	-0.009
IBD	y	Sparse logistic	0.726 (0.072)	0.669 (0.091)	-0.056
		XGBoost	0.794 (0.059)	0.765 (0.076)	-0.029
		MLP (base)	0.810 (0.047)	0.790 (0.070)	-0.020
	w_{class}	Sparse logistic	0.579 (0.027)	0.574 (0.029)	-0.005
		XGBoost	0.576 (0.019)	0.567 (0.012)	-0.009
		MLP (base)	0.607 (0.022)	0.605 (0.029)	-0.002
GvHD	y	Sparse logistic	0.625 (0.051)	0.620 (0.070)	-0.005
		XGBoost	0.782 (0.033)	0.780 (0.040)	-0.002
		MLP (base)	0.789 (0.039)	0.786 (0.043)	-0.003
	w_{class}	Sparse logistic	0.601 (0.013)	0.603 (0.015)	+0.002
		XGBoost	0.590 (0.015)	0.594 (0.018)	+0.005
		MLP (base)	0.596 (0.005)	0.595 (0.010)	-0.001

B Electronic appendix

The electronic appendix accompanies this thesis and contains everything needed to reproduce it. It is organised in three parts.

Code. The complete pipeline, as a set of independently runnable stages driven by a single configuration file, together with the R scripts that draw every figure, the static test suite, and a README describing the stage order and the outputs of each stage. The package versions and installation instructions are included with it.

Run outputs. The three cohort runs on which this thesis is based, one directory per cohort, each holding the frozen configuration used for that run, the preprocessed features, the fold assignments, the per-fold predictions and metrics, and the outputs of the ablation, the threshold sweep, the latent export and the shuffled- y control. The runs are 20260815_143422 (CRC), 20260815_174444 (IBD) and 20260815_202259 (GvHD); every figure and table in this thesis is drawn from these three.

Figures. All figures produced by the pipeline, including the descriptive and diagnostic views that are not reproduced in Section A because no statement in the text depends on them.

References

- Alrasheed, H., Jin, R. and Weitz, J. S. (2019). Caution in inferring viral strategies from abundance correlations in marine metagenomes, *Nature Communications* **10**: 501.
- Bengio, Y. and Grandvalet, Y. (2004). No unbiased estimator of the variance of k -fold cross-validation, *Journal of Machine Learning Research* **5**: 1089–1105.
- Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system, *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.
- Coenen, A. R. and Weitz, J. S. (2018). Limitations of correlation-based inference in complex virus-microbe communities, *mSystems* **3**(4): e00084–18.
- Coutinho, F. H., Zaragoza-Solas, A., López-Pérez, M., Barylski, J., Zielezinski, A., Dutilh, B. E., Edwards, R. A. and Rodriguez-Valera, F. (2021). RaFAH: Host prediction for viruses of bacteria and archaea based on protein content, *Patterns* **2**(7): 100274.
- Edwards, R. A., McNair, K., Faust, K., Raes, J. and Dutilh, B. E. (2016). Computational approaches to predict bacteriophage–host relationships, *FEMS Microbiology Reviews* **40**(2): 258–272.
- Friedman, J., Hastie, T. and Tibshirani, R. (2008). Sparse inverse covariance estimation with the graphical lasso, *Biostatistics* **9**(3): 432–441.
- Grinsztajn, L., Oyallon, E. and Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data?, *Advances in Neural Information Processing Systems 35*, pp. 507–520.
- Hille, F., Richter, H., Wong, S. P., Bratovič, M., Ressel, S. and Charpentier, E. (2018). The biology of CRISPR-Cas: Backward and forward, *Cell* **172**(6): 1239–1259.
- Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., Smetanin, N., Verkuil, R., Kabeli, O., Shmueli, Y., dos Santos Costa, A., Fazel-Zarandi, M., Sercu, T., Candido, S. and Rives, A. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model, *Science* **379**(6637): 1123–1130.
- Liu, H., Roeder, K. and Wasserman, L. (2010). Stability approach to regularization selection (StARS) for high dimensional graphical models, *Advances in Neural Information Processing Systems 23 (NIPS 2010)*, pp. 1432–1440.
- Lloyd-Price, J., Arze, C., Ananthakrishnan, A. N., Schirmer, M., Avila-Pacheco, J., Poon, T. W. and Huttenhower, C. (2019). Multi-omics of the gut microbial ecosystem in inflammatory bowel diseases, *Nature* **569**(7758): 655–662.
- Nie, W., Qiu, T., Wei, Y., Ding, H., Guo, Z. and Qiu, J. (2024). Advances in phage–host interaction prediction: in silico method enhances the development of phage therapies, *Briefings in Bioinformatics* **25**(3): bbae117.

- Peled, J. U., Gomes, A. L. C., Devlin, S. M., Perales, M.-A. and van den Brink, M. R. M. (2020). Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation, *New England Journal of Medicine* **382**(9): 822–834.
- Pinto, S., Benincà, E., van Nes, E. H., Scheffer, M. and Bogaards, J. A. (2022). Species abundance correlations carry limited information about microbial network interactions, *PLOS Computational Biology* **18**(9): e1010491.
- Roux, S., Camargo, A. P., Coutinho, F. H., Dabdoub, S. M., Dutilh, B. E., Nayfach, S. and Tritt, A. (2023). iPHoP: An integrated machine learning framework to maximize host prediction for metagenome-derived viruses of archaea and bacteria, *PLOS Biology* **21**(4): e3002083.
- Shang, J., Peng, C., Guan, J., Cai, D., Wang, D. and Sun, Y. (2025). From genomic signals to prediction tools: a critical feature analysis and rigorous benchmark for phage–host prediction, *Briefings in Bioinformatics* **26**(6): bbaf626.
- Shkoporov, A. N. and Hill, C. (2019). Bacteriophages of the human gut: The “known unknown” of the microbiome, *Cell Host & Microbe* **25**(2): 195–209.
- Steinegger, M. and Söding, J. (2017). MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets, *Nature Biotechnology* **35**(11): 1026–1028.
- Wirbel, J., Pyl, P. T., Kartal, E., Zych, K., Kashani, A., Milanese, A., Fleck, J. S., Bork, P. and Zeller, G. (2019). Meta-analysis of fecal metagenomes reveals global microbial signatures that are specific for colorectal cancer, *Nature Medicine* **25**(4): 679–689.
- Zhang, R., Mirdita, M., Levy Karin, E., Norroy, C., Galiez, C. and Söding, J. (2021). SpacePHARER: sensitive identification of phages from CRISPR spacers in prokaryotic hosts, *Bioinformatics* **37**(19): 3364–3366.

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I hereby confirm that I have written all parts of this text myself. I used Claude (Opus 4.8 and Opus 5, Cowork; Anthropic) as a tool: the source code of the analysis pipeline was generated with it under my direction, and I further used it for restructuring and refining the writing, for checking the analysis for errors, and for literature search. I specified the experimental design, the structure of the pipeline and every analytical choice, reviewed each stage and inspected its outputs. All suggestions were manually reviewed, adapted and reformulated, and I take full responsibility for the content of this work.

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